# VEERA HARISH MUTHAZHAGU

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## PROFESSIONAL SUMMARY

Full-Stack AI/ML Engineer with 3+ years of experience delivering enterprise-grade software and AI/ML solutions across the complete product lifecycle, from solution design and architecture through development, team leadership, deployment, release management, and production support. Proven track record of taking features from ground-up development through major production releases, with end-to-end ownership of maintenance, troubleshooting, bug resolution, and continuous improvement.

Strong expertise across Full-Stack Development, AI/ML, Generative AI, LLMs, RAG, and MLOps, with hands-on experience building scalable and production-ready solutions. Proficient in Python, JavaScript/TypeScript, Angular, React, Node.js, Express.js, SQL, MongoDB, FastAPI, MLflow, Kubernetes, Docker, and Azure.

Recognized for strong technical ownership, rapid adoption of new technologies, and delivering high-priority features within tight timelines. Experienced in leading teams, driving technical decisions, and managing critical releases across the SDLC. Published researcher in Artificial Intelligence and Machine Learning.

## CORE SKILLS

**Generative AI and LLM:** Generative AI, Large Language Models (LLMs), RAG, LangChain, Document Processing, Document Vectorization, Embeddings, Prompt Engineering, Multi-Source Data Integration

**AI/ML and MLOps:** Machine Learning, Deep Learning, MLOps, ML Lifecycle Management, Model Deployment, Model Inference, Model Monitoring, Model Registry, MLflow, Model Explainability, Time-Series Forecasting

**Full-Stack Development:** Python, JavaScript, TypeScript, Angular, React, Node.js, Express.js, FastAPI, REST APIs, Microservices, Async Programming, Pydantic

**Data Engineering and Analytics:** ETL, Data Processing, SQL, Microsoft SQL Server, MongoDB, Redis, Pandas, NumPy, Tableau

**Observability and Production:** OpenTelemetry, Prometheus, Grafana, Distributed Tracing, Metrics, Logging, Application Monitoring, Production Troubleshooting, Performance Optimization

**Cloud, DevOps and Orchestration:** Azure, Kubernetes, Docker, Helm, AKS, CI/CD, GitHub Actions, Control-M, Workflow Orchestration

**Architecture and Technical Leadership:** Cloud-Native Architecture, Distributed Systems, Event-Driven Architecture, Technical Leadership, Team Building and Mentoring, End-to-End Feature Ownership, Release Management, Production Support

## PROFESSIONAL EXPERIENCE

### Software Engineer / Tech Lead | ABB, Bengaluru | Aug 2024 - Present

Project: **ABB Ability Genix Model Fabric** - an enterprise MLOps platform for orchestrating and operating machine-learning workflows across cloud and edge environments, bringing together model registry, pipeline orchestration, recommendation, inference, and monitoring. Built with Python FastAPI/Celery microservices, Angular applications, a Go scheduler, and Kubernetes/Helm deployments.

- Progressed from individual contributor to Software Engineer and Feature Lead, taking ownership of complex enterprise features across AI/ML, GenAI, MLOps, and full-stack development.
- Designed and developed scalable AI/ML and GenAI platform capabilities using Python, FastAPI, Celery, MLflow, ONNX, RAG, LangChain, document vectorization, and multi-source data integration.

- Built model registry, pipeline orchestration, recommendation, inference, and monitoring capabilities for managing end-to-end machine-learning workflows across cloud and edge environments.
- Designed and implemented ML inference and model lifecycle workflows, covering model deployment, versioning, asynchronous processing, result handling, monitoring, and production troubleshooting.
- Led feature architecture and implementation across Angular frontends, FastAPI/Celery backends, a Go scheduler, ML, and Kubernetes/Helm deployment layers, collaborating with architects and cross-functional stakeholders.
- Built and mentored a development team from the ground up, providing knowledge transfer across architecture, workflows, deployment, and production environments to enable independent feature delivery.
- Owned major releases and production fixes, including root-cause analysis, code changes, validation, deployment, release coordination, and post-production monitoring.
- Rapidly adopted new frameworks and technologies based on evolving project requirements, delivering high-priority features within tight release timelines.

**Data Science Intern | ABB, Bengaluru | Sep 2023 - Jul 2024**

- Developed and evaluated machine learning solutions and data science workflows using Python.
- Worked on MLOps, model orchestration, deployment, and ML lifecycle workflows.
- Contributed to integrating data science solutions into enterprise applications and production workflows.

**Assistant System Engineer Trainee | Tata Consultancy Services, Kochi | May 2021 - Aug 2022**

- Supported the Analytics and Insights unit in enterprise reporting and data operations.
- Executed EDL jobs and data processing workflows, ensuring timely availability of data for reporting.
- Generated and maintained business reports using data visualization and analytics tools.
- Supported operational troubleshooting and resolution of issues across reporting workflows.

**React Native Developer Intern | Codesirpi Software LLP | Apr 2021 - May 2021**

- Developed cross-platform mobile application interfaces using React Native.
- Implemented UI components and enhancements focused on usability and user experience.

**Full Stack Developer Intern | ezPalate, Singapore | Jul 2020 - Sep 2020**

- Designed and developed an e-commerce food delivery platform using the MERN stack.
- Built frontend and backend components and integrated application functionality across the full development stack.

**Front-End Developer Intern | Steerclear Tech Solutions, Chennai | Apr 2020 - Jul 2020**

- Developed a B2B bulk-purchase portal for the fashion sector as part of the CAWACH initiative.
- Built and enhanced frontend components using the MERN stack, focusing on responsive UI and application usability.

## EDUCATION

Master of Technology (M.Tech), Computer Science and Engineering | National Institute of Technology Puducherry | 2022 - 2024 | CGPA: 9.32

Bachelor of Engineering (B.E), Computer Science and Engineering | Saranathan College of Engineering, Tiruchirappalli | 2017 - 2021 | CGPA: 8.61

## PROJECTS

### AI/ML and Software Projects

- **Waste Classification (SortGenie, CNN)** - Built a CNN-based solution for automated waste classification. Runner-Up (Pragyan24, NIT-T).
- **Voice-IDE (Python)** - Automated programming IDE that converts voice and keyboard input into generated code across multiple languages. Finalist (Pragyan21, NIT-T).
- **Transcoder (Java, Android)** - Real-time offline VoLTE call and message translation application. (CSI InApp19).
- **Deepfake Detection (EfficientNet-B4)** - Image-based deepfake detection system using EfficientNet-B4.

- **Fish Larvae Counting (R-CNN)** - Computer vision solution for automated fish larvae detection and counting using Region-based CNNs.
- **Air Quality Index Prediction (LSTM)** - LSTM-based time-series model for air quality index prediction.
- **Fish Disease Detection (MobileNetV2)** - Image classification model for fish disease detection using MobileNetV2.

### Company Hackathon Projects

- **Real-Time Model Governance (Guardrails)** - Real-time model governance concept implementing guardrails for responsible and controlled ML model usage. Runner-Up.
- **Proactive and Intelligent Model Management** - Intelligent model management solution combining Active Learning, AutoML, and Champion/Challenger deployment strategies. Finalist.
- **Fleet Federated Learning** - Federated learning system for industrial predictive maintenance, enabling distributed model training across fleet environments.
- **Failure-Aware Auto-Stabilization Engine** - Intelligent stabilization approach for detecting system failures and automatically initiating corrective actions to improve resilience.

## PUBLICATIONS

- Exploring the Role of AI in Web Design and Development: A Voyage Through Automated Code Generation, IITCEE 2024, DOI: 10.1109/IITCEE59897.2024.10467409.
- Automating Application Development by Processing User Commands, IRJMETS, Volume 4, Issue 3, March 2022.
- Deep Learning-Based Osteoarthritis Severity Classification from Radiographic Images (X-Ray), IEEE Xplore.
- Comparative Analysis of Mice Protein Expression Data: Assessing Genotype and Behavioral Treatments Using Machine Learning Algorithms, Springer.
- Navigating the AI Landscape: A Comparative Study of Models, Applications, and Emerging Trends, IEEE Xplore.

## CERTIFICATIONS

- Microsoft Technology Associate (MTA): Introduction to Programming using Python.
- IIT Bombay: Python Programming Fundamentals and Java Programming Fundamentals.
- Oracle: Java Fundamentals.
- HackerRank: Python Basics and React Basics.
- Udemy: React.js and Mastering Data Model Fundamentals.
- Learn Big Data: Hadoop Ecosystem Masterclass.
- BI Data Visualization: Tableau Intermediate.